
Detecting Credit Regimes in European Markets

A quantitative approach using Gaussian Mixture Models and lagged macro signals

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1. Motivation

European credit markets do not behave uniformly over time. Periods of compressed spreads and subdued volatility, characteristic of the quantitative easing era, alternate with periods of stress and dislocation driven by macro shocks. Identifying which regime the market is in - and detecting transitions before they fully reprice - is central to active credit portfolio management.

This note presents a systematic framework for detecting these regimes using publicly available data. The approach is motivated by the observation that the post-2022 environment represents a structural break from the decade of QE-suppressed volatility: central bank tightening, persistent inflation, and rising fiscal deficits have created conditions where tail risks in credit are underpriced relative to the macro backdrop. A regime detector that can identify these transitions early has direct implications for position sizing, hedging, and risk management in a liquid credit long/short strategy.

2. Data and Methodology

The analysis uses five daily time series from 2019 to 2024, sourced from the Federal Reserve Bank of St. Louis (FRED) and Yahoo Finance:

Variable	Description	Source
EUR HY OAS	Euro high yield option-adjusted spread (bps)	ICE BofA / FRED
EUR IG OAS	Euro investment grade OAS (bps)	ICE BofA / FRED
VIX	CBOE volatility index	FRED
US 10y yield	10-year Treasury yield (%)	FRED
EUR/USD	Euro/dollar exchange rate	FRED

Each series was tested for stationarity using the Augmented Dickey-Fuller (ADF) test. EUR HY OAS, EUR IG OAS, US 10y yield, and EUR/USD were found to be non-stationary in levels ($p > 0.05$) and were first-differenced prior to analysis. VIX was stationary in levels ($p = 0.0003$) due to its mean-reverting nature.

Regime detection was performed using a Gaussian Mixture Model (GMM) with three components, fitted on two features: the EUR HY OAS level and log-transformed VIX. Three components were chosen to represent risk-on, transition, and stress regimes. The model was initialised with 20 random seeds to avoid local optima, with the best solution selected by log-likelihood. States were labelled by their spread level mean - higher spread level corresponds to the stress regime.

Data limitation: The ICE BofA EUR HY series available on FRED provides reliable daily variation only from mid-2022 onwards. Prior to this the series is reported at lower frequency and forward-filled, which reduces the informativeness of daily spread changes in the earlier period. This is reflected in the regime output and is acknowledged as a constraint on the analysis.

3. Results

The GMM converged successfully and identified three well-separated regimes. The regime distribution across the 2019-2024 sample is as follows:

Regime	Days	Share	Avg Spread (bps)	Interpretation
Risk-on	1149	73.5%	Low / stable	QE-era compression, calm conditions
Transition	141	9.0%	Moving	Regime shift in progress
Stress	274	17.5%	Wide / volatile	Post-2022 repricing period

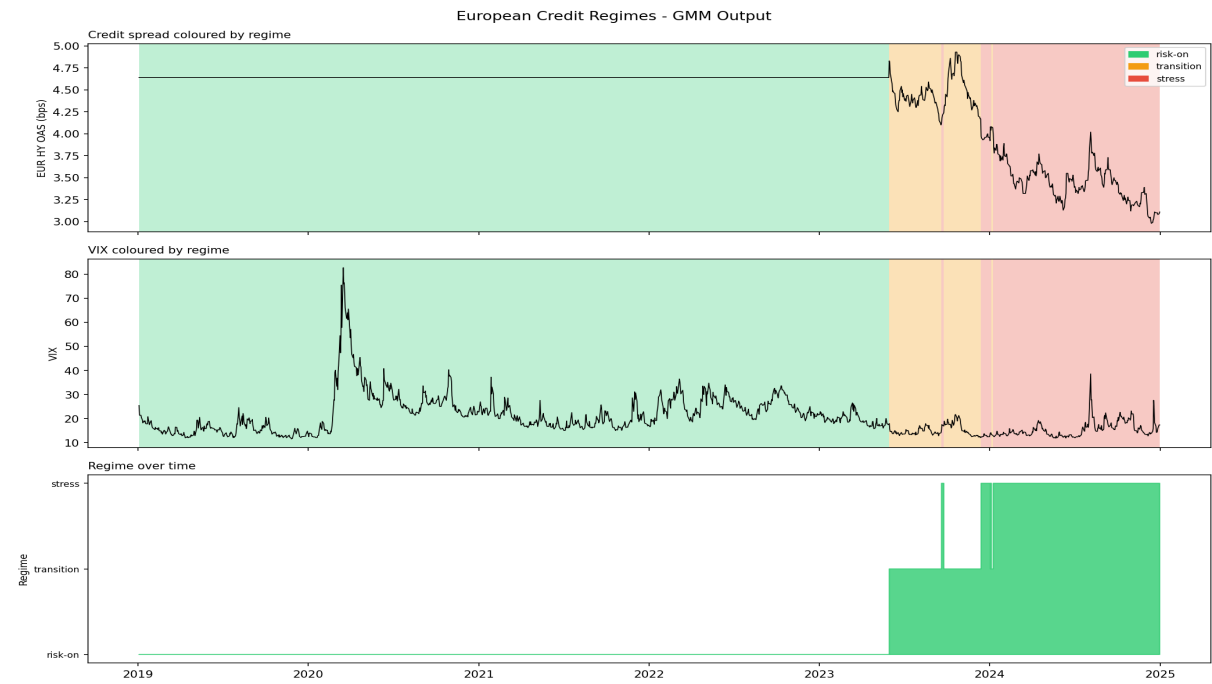


Figure 1. EUR HY OAS and VIX coloured by detected regime (green = risk-on, orange = transition, red = stress). The structural break in mid-2023 is clearly visible as the market shifts from the dominant risk-on regime.

4. Macro Leading Indicators

Having identified the regimes, the key analytical question is whether macro variables observed prior to a regime transition can predict it in advance. A logistic regression was fitted using lagged values of all five macro variables at horizons of 5, 10, and 20 trading days, with the GMM regime label as the dependent variable.

The key findings are:

- **Short lags dominate.** Predictive accuracy is highest at the 5-day lag (98.9%) and declines modestly at 10 days (98.6%) and 20 days (97.6%), suggesting regime signals are relatively short-lived and markets reprice quickly once a transition begins.
- **VIX leads risk-on regimes.** The 5 and 10-day lagged VIX are the strongest predictors of calm credit conditions, with coefficients of 1.77 and 1.42 respectively. Low volatility in the preceding week is a reliable signal that credit spreads will remain compressed.
- **Yield changes lead transitions.** The 5 and 10-day lagged yield is the strongest predictor of regime transitions, with coefficients of 1.40 and 1.27. This supports the view that rate moves precede credit regime shifts - a finding consistent with the post-2022 experience where aggressive central bank tightening drove the structural break in credit markets.
- **Spread persistence predicts stress.** Lagged HY and IG spreads are the strongest predictors of the stress regime, confirming that credit stress is persistent - wide spreads today predict wide spreads in the near term.

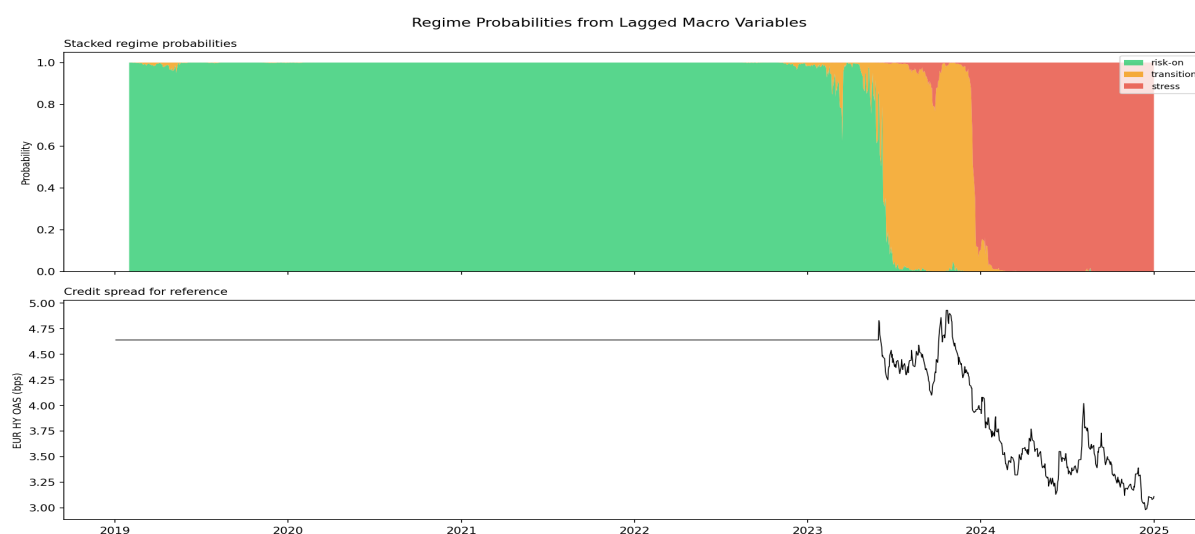


Figure 2. Stacked regime probabilities derived from lagged macro variables. The model assigns high confidence to risk-on through 2022, identifies a clear transition in mid-2023, and shifts to stress through 2024.

5. Implications and Limitations

The regime framework has direct applications for a liquid credit long/short portfolio. A manager observing a transition signal - rising yields over the preceding week alongside elevated VIX - could reduce gross exposure, increase short credit positions via iTraxx index hedges, or shift the portfolio toward higher-quality credits before spreads fully reprice. The 5-day lag finding suggests this signal provides approximately one trading week of lead time.

Several limitations constrain the current analysis. First, the in-sample accuracy figures overstate predictive power since the model is evaluated on the data it was trained on. A proper out-of-sample validation with a held-out test period would be the natural next step. Second, the data limitation in the EUR HY series prior to

mid-2022 means the risk-on regime is partly identified by the absence of daily variation rather than genuine spread compression signals. Access to proper iTraxx Main and Crossover daily data would significantly improve the pre-2022 regime identification. Third, the GMM does not impose temporal ordering on regimes - a Hidden Markov Model with transition probabilities would better capture the sequential nature of regime changes and reduce day-to-day switching noise.

Extensions to this framework could include incorporating PMI and ECB balance sheet data as additional macro factors, testing whether the regime signal adds value over a simple spread-level threshold rule, and applying the model in a rolling out-of-sample setting to simulate real-time regime detection.

Data sources: ICE BofA Indices via FRED (Federal Reserve Bank of St. Louis), CBOE VIX via FRED. All analysis conducted in Python using pandas, scikit-learn, and matplotlib. Code available at: [your GitHub link]